

Sales Prediction Analysis

B.Tech Machine Learning Project — B.I.T SINDRI



Student

Anuj Kumar | Roll No:
25SCS1003005282

Degree

B.Tech in Computer
Science and Engineering

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Organization

B.I.T SINDRI

Guide

Mr. Jitendra Kumar, Asst.
Professor, B.I.T Sindri

University

IILM University, Greater Noida, Uttar Pradesh

Executive Summary

A high-level overview of the project's objectives, key findings, and impactful results.



Project Objective

Developed a robust machine learning model to accurately forecast sales, enhancing operational efficiency.



Key Finding

Analysis revealed revenue predictability significantly surpasses profit predictability due to diverse market dynamics.



Model Performance

Achieved an R^2 score of **0.860** with the optimized XGBoost model, indicating high predictive accuracy.



Business Impact

Provided actionable insights for improved inventory management, targeted marketing, and data-driven strategic planning.

Project Overview

A comprehensive machine learning pipeline designed to predict **Revenue and Profit** from real-world retail order data.

Dataset Scale

10,000 orders spanning January 2023 – December 2024

Product Coverage

4 categories, 19 sub-categories, 49 distinct products

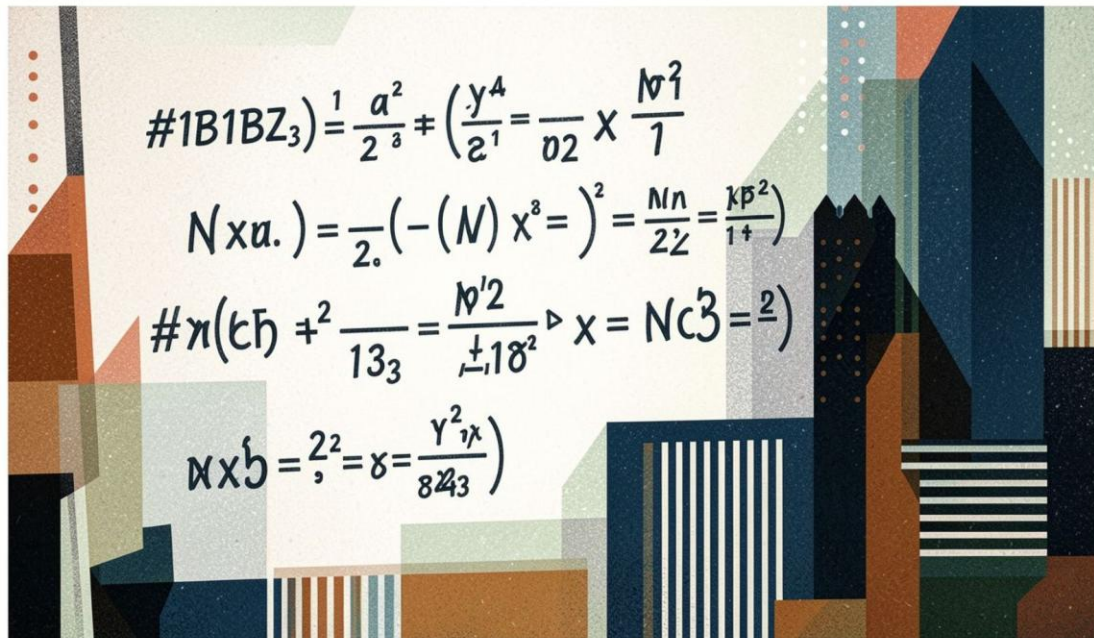
Geographic Reach

4 distinct regional markets analysed

Objective

Develop regression models for accurate sales forecasting

Key Finding: Revenue Formula


$$\begin{aligned} \#1B1BZ_3) &= \frac{1}{2} \frac{a^2}{2} \neq \left(\frac{y^4}{2^1} = \frac{1}{2} \times \frac{10^2}{1} \right. \\ N(xa.) &= \frac{1}{2} \left(- (N) x^3 = \right)^2 = \frac{Mn}{2Z} = \frac{kp^2}{1+} \\ \#n(ck) &+^2 \frac{1}{13_3} = \frac{10^2}{18^2} \triangleright x = Nc3 = \frac{2}{2} \\ Nx6 &= \frac{2^2}{5} = 8 = \frac{Y^2, x}{843} \end{aligned}$$

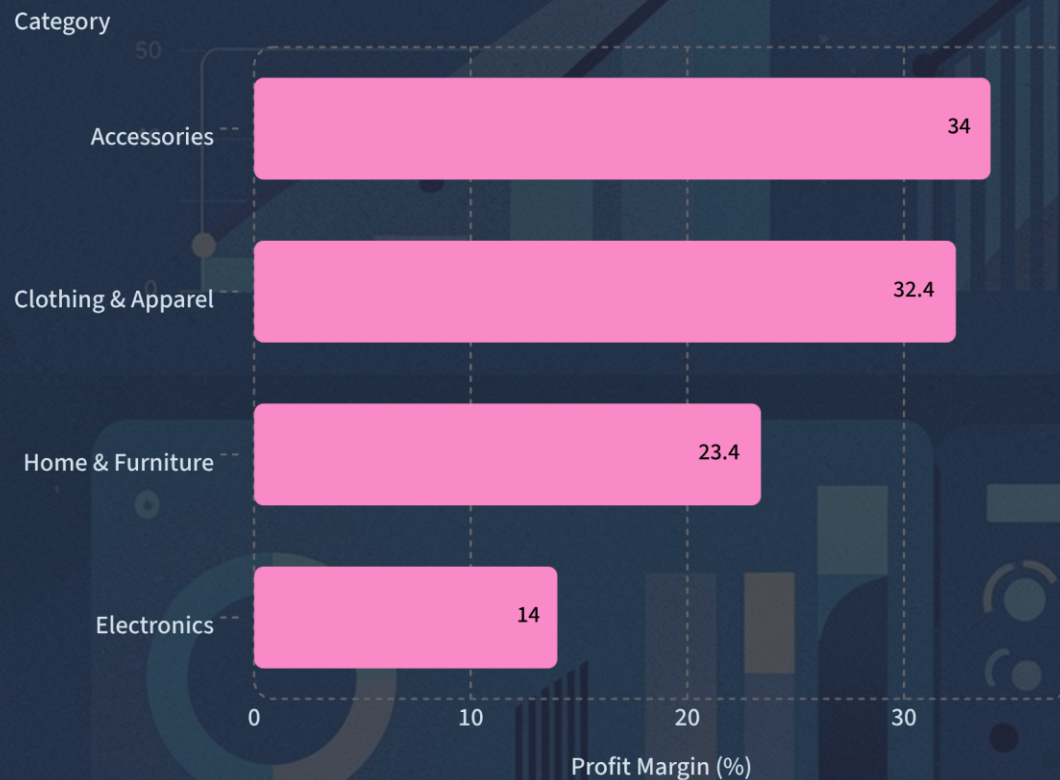
A Critical ML Insight

Revenue follows an **exact deterministic formula** — it is arithmetic, not a learnable relationship.

❏ **Revenue = Quantity × Unit Price** —
This identity means ML models cannot add value here.

- Quantity and Unit Price are **excluded** from the Revenue model to preserve ML integrity
- Profit identified as a **genuinely learnable target**
- Strong correlation: $r = 0.84$ between Profit and Revenue

Data Insights



What the Data Reveals

- Accessories lead profit margins at **34.0%**
- Electronics trail significantly at just **14.0%**
- Both Revenue and Profit show **right-skewed distributions**
- Strong **seasonal peaks in November** across both years
- Time-based train/test split: **83% / 17%** to prevent temporal data leakage

Modelling Approaches

Three regression techniques were evaluated — from a simple linear baseline to powerful ensemble methods.



Linear Regression

Baseline model establishing minimum performance benchmark



Random Forest

Ensemble of decision trees — **best performer** across both targets



XGBoost

Gradient boosting ensemble used for rigorous comparison

 Revenue models scored $R^2 = 0.14\text{--}0.26$, confirming non-learnability. Profit models saw tree-based methods substantially outperform the linear baseline.

Best Model Performance

Tuned Random Forest Regressor on the Profit target — the standout model of this project.

0.860

R^2 Score

On held-out test data

\$36.35

MAE

Mean Absolute Error

\$56.55

RMSE

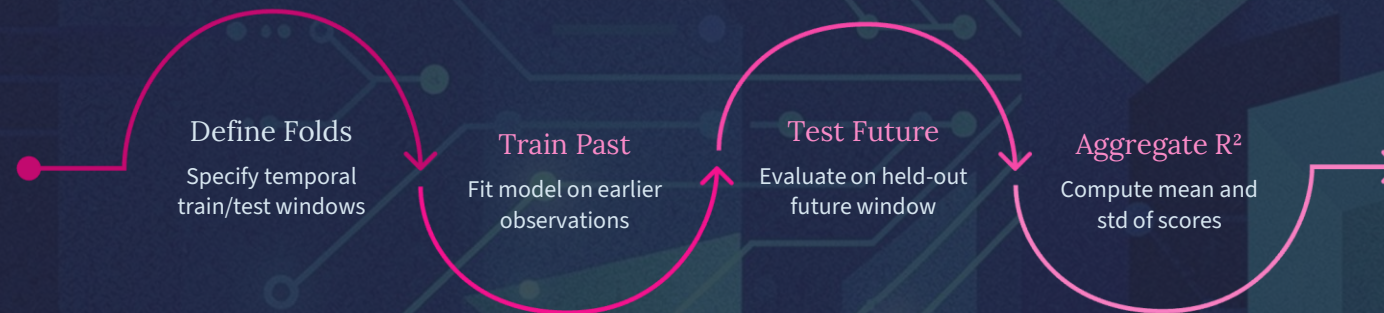
Root Mean Square Error

+6.5pp

Improvement

Over linear baseline

Model Validation



Rigorous Temporal Testing

- **Five-fold time-series cross-validation** conducted to prevent look-ahead bias
- Mean $R^2 = \mathbf{0.847}$ with a standard deviation of just **0.016**
- Confirms **stable generalisation** across temporal windows
- Hyperparameter tuning via Randomised Search CV yielded **negligible improvement**
- Model already near-optimally configured for the dataset scale

Feature Importance: SHAP Analysis

What Drives Profit Predictions?

SHAP values provide direct, actionable insight into how each feature influences the model's output — making the model transparent and trustworthy for business stakeholders.

#SHAP



Revenue

Strongest positive predictor of Profit

Category

Electronics reduces predicted profit by ~\$66 vs dataset mean

Quantity & Unit Price

Key secondary features driving model decisions

Conclusion

This project demonstrates that machine learning can deliver reliable, interpretable profit predictions from categorical and temporal features alone.

1 Revenue Forecasting Requires Order-Size Inputs

Quantity and Unit Price are essential inputs for Revenue — without them, the target collapses to arithmetic identity.

2 Profit Is Reliably Predictable

Random Forest achieves $R^2 = 0.860$ using only categorical and temporal features — no order-size leakage required.

3 Actionable Business Intelligence

SHAP-validated insights support data-driven strategies for inventory planning and margin optimisation across all product categories.

✓ The Random Forest model is production-ready for business decision support in sales forecasting and margin management.